

Contraction Theory — Derivation Notes

(Note 20 Feb 2026)

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March 2026

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1 Setup

Consider a smooth nonlinear system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t), \quad \mathbf{x} \in \mathbb{R}^n. \quad (1)$$

Let $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ be *two solutions* of (1) starting from different initial conditions (ICs). Define the **virtual displacement**

$$\delta \mathbf{x} = \mathbf{x}_1 - \mathbf{x}_2 \in \mathbb{R}^n,$$

which satisfies the **variational dynamics** (linearised along any trajectory):

$$\delta \dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, t) \delta \mathbf{x}, \quad \mathbf{F} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}. \quad (2)$$

2 Metric and Transformed Displacement

Introduce a uniformly positive-definite **metric** (coordinate transformation)

$$\Theta(\mathbf{x}, t) \in \mathbb{R}^{n \times n}, \quad \Theta^\top \Theta = \mathbf{M}(\mathbf{x}, t) \succ 0,$$

and define the **transformed virtual displacement**

$$\delta \mathbf{y} = \Theta \delta \mathbf{x}. \quad (3)$$

The **squared length** (Lyapunov-like measure) is

$$V = \delta \mathbf{y}^\top \delta \mathbf{y} = \delta \mathbf{x}^\top \mathbf{M} \delta \mathbf{x} \geq 0. \quad (4)$$

3 Time Derivative of Squared Length

Differentiating (4) along (2):

$$\begin{aligned} \dot{V} &= \delta \dot{\mathbf{y}}^\top \delta \mathbf{y} + \delta \mathbf{y}^\top \delta \dot{\mathbf{y}} \\ &= 2 \delta \mathbf{y}^\top \delta \dot{\mathbf{y}}. \end{aligned} \quad (5)$$

From (3): $\delta \dot{\mathbf{y}} = \dot{\Theta} \delta \mathbf{x} + \Theta \delta \dot{\mathbf{x}} = (\dot{\Theta} + \Theta \mathbf{F}) \delta \mathbf{x} = (\dot{\Theta} + \Theta \mathbf{F}) \Theta^{-1} \delta \mathbf{y}$.

Define the **generalised Jacobian**

$$\mathbf{K}(\mathbf{x}, t) = (\dot{\Theta} + \Theta \mathbf{F}) \Theta^{-1}, \quad (6)$$

so that $\delta \dot{\mathbf{y}} = \mathbf{K} \delta \mathbf{y}$, and (5) becomes

$$\boxed{\dot{V} = 2 \delta \mathbf{y}^\top \mathbf{K} \delta \mathbf{y} = \delta \mathbf{y}^\top (\mathbf{K} + \mathbf{K}^\top) \delta \mathbf{y}.} \quad (7)$$

4 Contraction Condition

Theorem 1 (Contraction). *The system (1) is **contracting** with respect to the metric Θ if there exists a constant $\lambda > 0$ such that*

$$\mathbf{K}_s \triangleq \frac{1}{2}(\mathbf{K} + \mathbf{K}^\top) \preceq -\lambda \mathbf{I}, \quad \forall \mathbf{x}, t. \quad (8)$$

That is, the symmetric part of $\mathbf{K}$ is uniformly negative definite.

Proof. From (7) and (8):

$$\dot{V} \leq -2\lambda V.$$

By Grönwall’s lemma:

$$V(t) \leq V(0) e^{-2\lambda t},$$

i.e.

$$\|\delta \mathbf{y}(t)\| \leq \|\delta \mathbf{y}(0)\| e^{-\lambda t}. \quad (9)$$

Since Θ is uniformly bounded and invertible, this gives **exponential convergence** of all trajectories to each other:

$$\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\| \rightarrow 0 \quad \text{exponentially.}$$

□

5 Special Case: Identity Metric

With $\Theta = \mathbf{I}$ the generalised Jacobian reduces to the ordinary Jacobian $\mathbf{K} = \mathbf{F}$, and the contraction condition becomes:

$$\mathbf{F}_s = \frac{1}{2}(\mathbf{F} + \mathbf{F}^\top) \preceq -\lambda \mathbf{I}. \quad (10)$$

This is the *Euclidean* contraction condition.

6 Examples

Example (a) — Linear time-invariant system

Let $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, $\mathbf{A} \in \mathbb{R}^{n \times n}$ constant. With $\Theta = \mathbf{S}$ (constant), $\mathbf{K} = \mathbf{S}\mathbf{A}\mathbf{S}^{-1}$. Choose $\mathbf{S}^\top \mathbf{S} = \mathbf{M}$ such that $\mathbf{K} + \mathbf{K}^\top \prec 0$:

$$\mathbf{S}\mathbf{A}\mathbf{S}^{-1} + \mathbf{S}^{-\top} \mathbf{A}^\top \mathbf{S}^\top \prec 0 \iff \mathbf{A}^\top \mathbf{M} + \mathbf{M}\mathbf{A} \prec 0 \quad (\mathbf{M} = \mathbf{S}^\top \mathbf{S}).$$

This is the standard Lyapunov stability condition — contraction subsumes it.

Example (b) — Scalar nonlinear system

Let $\dot{x} = f(x) = -x - x^3$. Then $F = \partial f / \partial x = -1 - 3x^2$. For all x , with $\Theta = 1$:

$$K_s = F = -1 - 3x^2 \leq -1 < 0.$$

So $\lambda = 1$; the system is globally contracting and all trajectories converge exponentially.

7 Summary

Quantity	Definition / Role
$\delta \mathbf{x}$	Virtual displacement between two trajectories
Θ	Uniformly invertible coordinate transformation (metric)
$\delta \mathbf{y} = \Theta \delta \mathbf{x}$	Transformed displacement
$V = \ \delta \mathbf{y}\ ^2$	Squared length (Lyapunov measure)
$\mathbf{K} = (\dot{\Theta} + \Theta \mathbf{F})\Theta^{-1}$	Generalised Jacobian
$\mathbf{K}_s \preceq -\lambda \mathbf{I}$	Contraction condition ($\lambda > 0$)
$\ \delta \mathbf{x}\ \rightarrow 0$ exponentially	Conclusion (all-to-all convergence)

Key insight: A system is contracting if a metric Θ can be found such that the symmetric part of the generalised Jacobian $\mathbf{K}$ is uniformly negative definite. Unlike Lyapunov methods, this gives *any-to-any* trajectory convergence, not just stability of a single equilibrium — making it directly applicable to the cup-and-ball transport task studied in Bazzi et al. (2018).